Auditing Normal-Reference Coordinate Drift in a Staged Skeleton JEPA

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Abstract

Continual representation learning can change how a model encodes normal-labeled reference rows. We audit this change in a project-specific, S-JEPA-inspired skeleton model trained first on normal-labeled gait and then on four condition-labeled groups with balanced replay. The optional added-normal cohort is off. After availability and pose-quality checks, the experiment contains 626 sequences from 93 source videos. For each of 270 normal training sequences, we compare its current representation with its own Stage-0 representation and average the cosines. This raw normal-anchor cosine falls from 0.700 after the first later stage to 0.297 after the fourth; checkpoint reload reproduces the curve within 4.51×10^{-7}. This is coordinate drift, not proof of forgetting: raw cosine is basis-dependent, the reference rows were used in training, the anchor is source-imbalanced, and the run has one seed. Sequence-split probes reach 0.899 macro-F1, but all test rows were encoder-exposed and 64 source videos cross the classifier split. We present an auditable measurement protocol and the controls needed for functional-retention and generalization claims. This is not a diagnostic or clinical-validation study.

1 Introduction

Models of movement often learn a reference and then adapt to new tasks or populations. If that reference moves, later comparisons to "normal" also change. A simple audit saves the representation of every normal sequence after initial training and compares the matched representations after each later stage.

That audit needs careful interpretation. Neural features do not have a fixed coordinate system. Two encoders can retain the same relationships in rotated bases while their raw matched-vector cosine falls. Raw drift therefore does not establish functional forgetting. Forgetting needs aligned representation comparisons and worse performance on held-out normal data.

We study this distinction in a normal-first gait model. The model is inspired by S-JEPA, but it is not a reproduction of the published method: it uses a fixed 12-landmark target sampler, VICReg, and a label-aware group term after Stage 0. The experiment exposes common evaluation risks: source-video overlap, encoder exposure to evaluation rows, pose-detector shortcuts, stale artifact lineage, and one-seed conclusions.

Our contributions are: (1) a checkpoint-recomputed raw normal-anchor curve; (2) a claim audit that separates coordinate drift, functional retention, in-corpus label readability, and out-of-source generalization; and (3) a minimum experiment set for a stronger continual-learning claim.

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2 Related work

JEPAs predict target-encoder features rather than reconstructing every input value. I-JEPA applies this idea to images [2], V-JEPA to video [4], and S-JEPA to skeleton sequences [1]. VICReg discourages collapse through invariance, variance, and covariance terms [3]. Our study is not an action-recognition benchmark or clinical classifier. It audits a continually updated representation and asks which conclusions its artifacts support.

3 Method

3.1 Data and pose processing

GAVD supplies video-linked rows with one of five folder annotations: normal, Parkinson's, stroke, myopathic, or cerebral palsy [8]. These are dataset annotations, not diagnoses made by this project. The raw local inventory contains 666 sequences from 103 source videos. Availability checks retain 642 sequences from 94 videos. Requiring at least 0.50 observed fraction over the 12 gait-focused target landmarks removes 16 rows and leaves Table 1.

Table 1: Current GAVD-only modeled cohort.

Annotation	Sequences	Source videos
Normal	270	29
Parkinson's	41	9
Stroke	74	18
Myopathic	183	28
Cerebral palsy	58	9
Total	626	93

The availability-filtered pose caches have mixed extraction labels: 546 `gavd5`, 95 `gavd3`, and one `gavd4`; the modeled cohort has 530, 95, and one. The recorded pose-model hash agrees, but extraction history may still correlate with source or label. GAVD has no reliable person identifier for this analysis, so source-video grouping cannot guarantee person separation.

MediaPipe BlazePose produces 33 landmarks and visibility values per frame [6, 7]. We interpolate short internal gaps, center each frame on the pelvis, normalize by body scale, and resize time to 64 frames. A token represents one landmark over four frames. Missing or low-visibility tokens cannot become prediction targets. Targets are sampled uniformly from shoulders, hips, knees, ankles, heels, and foot tips. The configured 0.60 mask fraction is applied to the smallest eligible-token count in the batch.

3.2 Model and curriculum

The model has a view encoder, an exponential-moving-average target encoder, and a predictor. Encoder width is 96, with four encoder blocks and two predictor blocks. The objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{JEPA}} + 0.05\mathcal{L}_{\mathrm{VICReg}} + 0.25\mathcal{L}_{\mathrm{group}}. \tag{1}$$

The group term is zero during normal-only Stage 0. Later it uses folder labels to encourage within-condition compactness and a margin between condition centroids. Later training is therefore label-informed, not purely self-supervised.

One model continues across all stages. Stage 0 trains for 300 epochs on 270 normal rows. Four 75-epoch stages add the other annotations in the order shown in Table 1, with four rows per active condition in each balanced-replay batch. Model, EMA target, center, and projector state continue; AdamW restarts at every stage. AdamW uses betas $(0.9, 0.95)$, weight decay 0.05, learning rate 10^{-3} at Stage 0, and 3×10^{-4} later. The run has 600 epochs, 40,800 optimizer updates, seed 42, and the MPS backend. Exact hardware and full deterministic settings were not recorded.

3.3 Normal anchor and claim levels

Let $z_0(x) \in \mathbb{R}^{96}$ be the Stage-0 validity-weighted mean of target-encoder tokens at the 12 authorized landmarks for normal sequence x. After stage t, we report

$$a_t = \frac{1}{|N|} \sum_{x \in N} \frac{z_t(x)^\top z_0(x)}{\|z_t(x)\|_2 \|z_0(x)\|_2}. \tag{2}$$

This is an average of matched per-sequence cosines, not a cosine between cohort centroids. It uses known normal labels and encoder-exposed training rows. It also weights sequences, not videos, equally: the top two normal videos supply 105 of 270 rows and the top three supply 137. We therefore treat it as training-corpus telemetry.

4 Results

4.1 Raw coordinates change across the curriculum

The anchor cosines after the four later stages are 0.7002, 0.5021, 0.3962, and 0.2966 (Figure 1). Reloading the five checkpoints gives a maximum absolute difference of 4.51×10^{-7} from the saved summary. This rules out a simple logging error. It does not establish catastrophic forgetting because the run lacks basis alignment, a continued-normal control, alternate condition order, held-out normal function, and multiple seeds.

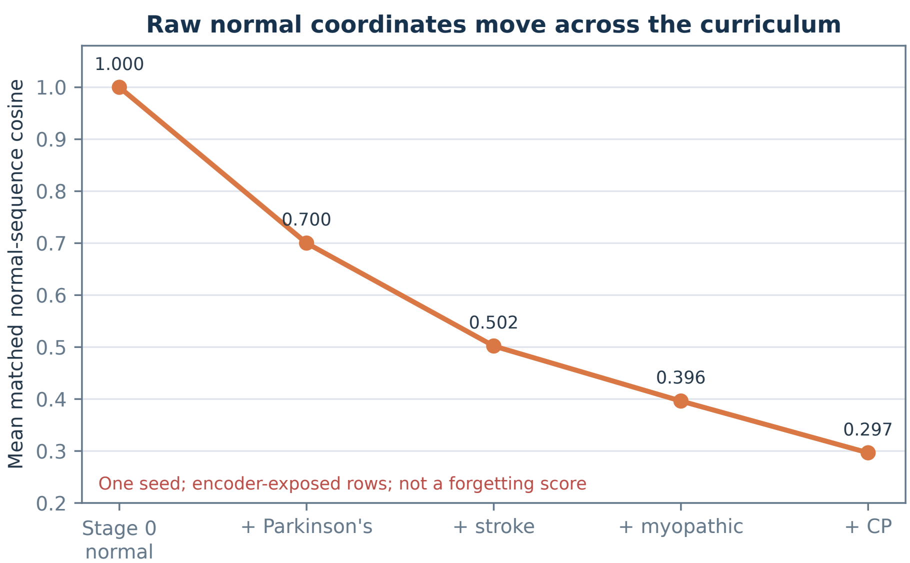

Figure 1: Mean matched-sequence normal-anchor cosine. Stage 0 defines the reference at 1.0. This is a raw coordinate measure for one seed, not a forgetting score.

4.2 The final representation is structured in-corpus

At Stage 4, the 96-D authorized target-token means have feature standard deviation 0.229 and mean pair cosine similarity 0.440. Their smallest condition-centroid distance is 0.534, measured as Euclidean distance between normalized centroids. A separate frozen analysis concatenates global mean, global standard deviation, authorized-landmark mean, and authorized-landmark standard deviation into a 384-D vector. In that space, cosine silhouette is 0.3617, mean within-condition cosine distance is 0.0783, and minimum between-centroid cosine distance is 0.0863. These values rule out total constant collapse and show label-informed in-corpus structure; they do not validate an invariant clinical representation.

4.3 Classifier scores are descriptive readouts

We fit Random Forest readouts to the frozen 384-D summaries. Table 2 reports the two current sequence splits and a missingness-only control.

Table 2: Five-class frozen-feature probes.

Probe	Accuracy	Bal. acc.	Macro-F1
All 626 rows, sequence split	0.9202	0.9001	0.8985
Exact historical 47/21 split	0.8571	0.8800	0.8607
Missingness only, all rows	0.4415	0.4269	0.3547

In the all-row split, all 188 classifier test rows were already used by the encoder, 64 source videos appear on both sides, and 181 of 188 test rows come from shared videos. In the exact split, all 9 test videos overlap classifier training. These scores show that fitted features encode in-corpus folder labels; they do not estimate new-video or new-person performance. Grouped validation is required when rows share a source [9]. Detector visibility alone also contains label signal, although its macro-F1 is lower than the model readout.

5 Discussion

The current run establishes a substantial raw-coordinate change: mean matched-sequence cosine reaches 0.297. That observation is useful telemetry. It does not establish lost function. The representation may rotate, continued optimization alone may move it, or the curve may depend on seed 42 and the fixed condition order.

An adversarial artifact audit excluded stale or mixed-lineage margin-ablation, AnchorGuard, grouped-classifier, and forecasting results. We therefore make no mechanism, repair, forecasting, or generalization claim. The executed temporal and signed-laterality ridge probes also emit many numerical ill-conditioning warnings and lack repeated-split intervals; neither clears its pre-specified success gates (Appendix D).

The work has broader data limits. A source video is not a person. Folder labels are not independently adjudicated diagnoses. Camera, compression, framing, pose missingness, and extraction history may correlate with labels. Multiple training seeds would not solve the anchor's source pseudoreplication. No result supports clinical use.

Data use and ethics. GAVD distributes annotations and public video URLs rather than raw video files; users retrieve media independently and must comply with YouTube terms, institutional ethics requirements, and applicable copyright, privacy, and data-protection rules [5]. This analysis uses derived pose sequences and does not infer identity. A public artifact should not redistribute raw videos or identity-bearing frames. The institutional ethics determination and data-use review must be recorded before submission.

6 Experiments required for strong claims

The minimum P0 set is three to five full-curriculum seeds; per-video and equal-video-weighted anchors with source-cluster uncertainty; Procrustes plus CKA or SVCCA; a same-update continued-normal control; and retraining with a source-grouped normal holdout. A condition-order control is also needed. Mechanism claims need matched group-loss ablations; generalization needs preprocessing, encoder training, and readout fitting inside every outer source-video fold. Appendix C gives the complete map.

7 Conclusion

Raw coordinate drift, functional retention, and clinical performance are distinct. This one-seed GAVD-only run supports only the first and shows that high in-corpus decoding can coexist with drift; it does not support forgetting, generalization, or clinical use.

A Reproducibility ledger

- Cohort: 626 sequences, 93 videos; counts 270 / 41 / 74 / 183 / 58.
- Dataset fingerprint: `7d13841aceac9eda843d43ca8434193e294d2fa10a48b6c6d21f6413a6e457e2`.
- Checkpoint SHA-256: `64008d77689cefa4beb51a0dcf5ed6cae743454134c163e9087f66510af4e7ad`.
- Anchor curve: 0.700151 / 0.502113 / 0.396213 / 0.296638; reload gap at most 4.51×10^{-7}.
- 384-D geometry: silhouette 0.361717; minimum between 0.086261; within 0.078339.
- All-row readout: accuracy 0.920213; macro-F1 0.898512. Missingness control: accuracy 0.441489; macro-F1 0.354682.

B Excluded artifact families

The Stage-1 margin ablation loads an old augmented Stage-0 checkpoint and fails to reproduce the current Stage-1 endpoint. The cached AnchorGuard result lacks complete current parent and cohort lineage. The grouped "Lane C" file names the old augmented encoder and 159-row cohort. The forecasting notebook combines that old checkpoint with current rows. The AnchorGuard non-inferiority gate also uses an absolute difference where a one-sided rule is required. None of these results supports the present claims.

C Experiment-to-claim map

- **Repeatability:** three to five full seeds and alternate or randomized order.
- **Independent unit:** per-video anchors, equal-video weighting, source-cluster uncertainty, and source-balanced replay.
- **Basis invariance:** orthogonal Procrustes plus linear CKA or SVCCA.
- **Optimization control:** continued-normal and joint-training controls with matched updates.
- **Function:** source-grouped normal holdout, retraining, and checkpoint-wise JEPA loss and perturbation ranking.
- **Mechanism and repair:** current-lineage group-loss and AnchorGuard studies with identical random streams and one-sided gates.
- **Probe stability:** stable SVD-based solvers, regularization sensitivity, label permutation, and repeated grouped splits.
- **Generalization:** fold-local preprocessing, encoder training, and readout fitting.

D Secondary probe details

The pre-specified temporal-moment lane requires at least 10% lower pooled error and improvement in at least 75% of source videos. Relative pooled improvements are 4.0%, -1.9%, and 9.2%; no target clears both gates. Target counts range from 539 to 626. For signed laterality, learned tokens reach $R^2 = 0.241$ versus 0.190 for an untrained encoder, but sign consistency is 55.3% and mirror slope is -0.627, outside the required $[-1.25, -0.80]$. The raw-feature score near 1 is a target-construction sanity ceiling, not validation. The laterality probe uses the broader 642-row cohort. Both studies need stable solvers and repeated grouped splits.

References

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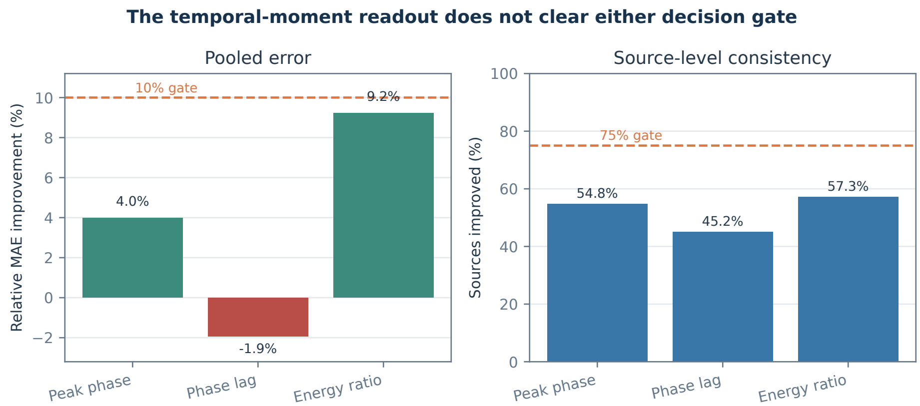

Figure 2: The pre-specified temporal-moment readout does not clear either decision gate.

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